PID-RS TECHNICAL REPORT SERIES

Finite-Alphabet Plug-In Convergence for Shared-Exclusions PID

Exact-real theorem, local moduli, an anytime i.i.d. envelope, and a formal evidence boundary

pid-rs project analysis

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Abstract

This document gives new project-defined validation for existing information measures in `pid-rs`. Its primary object is the Wibral-lineage categorical shared-exclusions partial information decomposition (SxPID). The paper-defined functional is not new. The document proves exact-real plug-in convergence on fixed finite alphabets, gives explicit local continuity moduli, and composes them with a conservative time-uniform i.i.d. concentration event. A secondary corollary covers the Williams–Beer $I_{\min}$ functional and fixed Shannon combinations. A frozen-transform corollary states explicit sufficient conditions under which an independently trained finite map preserves the argument. The theorem does not prove convergence of binary64 program outputs, dependent-window validity, drift calibration, or scientific novelty. Invalid stronger claims and counterexamples remain in the document.

A separate repository paper on SxPID under a dependency coloring gives a categorical result for a predeclared independence coloring. Its Theorem DC-1 separately assumes $\mathrm{Law}(Z_i) = P$ for every complete row and mutual independence inside every color class; identical distribution and independence are not conflated. Theorem DC-3 handles nonidentical-law drift. Those results do not change the i.i.d. and ergodic claims in this paper, and they do not cover generic mixing, outcome-adaptive colors, or continuous estimators.

Contents

1 Status and provenance

The mathematical object and the validation artifact have different origins.

Object	Origin	Result in this document
Categorical SxPID	Paper-defined by Makkeh, Gutknecht, and Wibral; its part-whole lattice has a formal-logic basis due to Gutknecht, Wibral, and Makkeh [7, 4]	Exact-real convergence for two to four sources, local moduli, and bounded code evidence
I_{min}	Paper-defined by Williams and Beer [12]	Secondary finite-minimum corollary for two and three sources
Shannon entropy and mutual information	Paper-defined classical quantities [11]	Exact-real convergence and entropy continuity bounds
Discrete co-information and O-information	Paper-defined quantities [2, 8, 9]	Convergence as fixed finite entropy combinations
Target-free normalized ratios	Project-defined analogues in `pid-rs`	Positive-denominator ratio convergence
Target-conditioned normalized ratios	Paper-defined ratios with a project-defined report policy [5]	Positive-denominator ratio convergence and a status boundary
Fitted quantizer plus categorical PID	Project-defined composition; quantization has published background [3]	Conditional frozen-transform corollary

The machine-readable authority is `method-catalog.json`. The rendered authority is `METHODS.md`. This document defines no new estimator or public API. The exact Rust routes covered by the result are:

- SxPID namespace `stable::categorical`: `discrete_sxpid2`, `discrete_sxpid2_averaged`, `discrete_sxpid3`, `discrete_sxpid3_averaged`, `discrete_sxpid_n`, and `discrete_sxpid_n_averaged`;
- I_{min} namespace `stable::imin`: `imin_pid2` and `imin_pid3`;
- Shannon namespace `diagnostics`: `entropy_discrete`, `joint_entropy_discrete`, `co_information_pairwise_discrete`, and `o_information_discrete`;

- normalized-report namespace `diagnostics`: `red_degree_discrete`, `vul_degree_discrete`, `average_degree_of_redundancy`, and `average_degree_of_vulnerability`;
- fitted SxPID namespace `stable::quantized`: `fitted_quantized_sxpid2`, `fitted_quantized_sxpid3`, and `fitted_quantized_sxpid_n`;
- fitted $I_{\min}$ namespace `stable::imin`: `imin_pid2_quantized` and `imin_pid3_quantized`.

What problem this note solves

An implementation can evaluate a published categorical functional at an empirical law, but code existence alone does not prove that the plug-in value approaches a population value. The key mathematical chain is

$$\begin{aligned} \widehat{P}_n \to P \implies& \text{ event masses converge and support stabilizes} \\ \implies& \text{ required logarithms are eventually defined} \\ \implies& \text{ finite Möbius inversion and averaging preserve convergence.} \end{aligned} \tag{1}$$

FA-1 proves this chain deterministically. FA-2 supplies two standard stochastic routes to its premises. FA-3 adds a conservative time-uniform i.i.d. radius. Keeping these layers separate prevents a strong law, a finite-sample bound, a PID identity, and binary64 behavior from being mistaken for one another.

Reader's guide, foundational notation, and prerequisites

No prior knowledge of PID is required for the arguments below. A *probability law* Q on a finite set $\mathcal{Z}$ is a vector $(Q(z))_{z\in\mathcal{Z}}$ with nonnegative entries summing to one. Its support is

$$\operatorname{supp}(Q) = \{z \in \mathcal{Z} : Q(z) > 0\}.$$

For two such laws,

$$\|Q-P\|_1 = \sum_{z\in\mathcal{Z}} |Q(z)-P(z)|, \qquad \mathrm{TV}(Q,P) = \frac{1}{2}\|Q-P\|_1 = \sup_{E\subseteq\mathcal{Z}} |Q(E)-P(E)|.$$

Marginalization means summing joint-cell masses over coordinates that are not retained; it cannot increase L^1 distance. A *plug-in* quantity is the population formula evaluated after replacing the unknown law by an empirical law.

PID seeks coordinates whose sum reconstructs a mutual-information quantity while distinguishing information shared by several sources, information unique to one source, and information available only jointly. SxPID constructs these coordinates from logical source events. Put $[m] = \{1,\dots,m\}$. A nonempty set $a \subseteq [m]$ is a *source collection*; $S_a = (S_i)_{i\in a}$ and $s_a = (s_i)_{i\in a}$. Thus

$$\{S_a = s_a\} = \bigcap_{i\in a}\{S_i = s_i\}.$$

A lattice node α is a nonempty *antichain* of nonempty source collections: no two distinct members of α contain one another. The redundancy order used here is

$$\alpha \preceq \beta \iff \text{for every } b\in\beta \text{ there is an } a\in\alpha \text{ with } a\subseteq b.$$

For $u \in \{+,-,\mathrm{sx}\}$, cumulative coordinates and pointwise atoms are related by the finite poset zeta transform

$$c^u_\alpha(z;Q) = \sum_{\beta\preceq\alpha} \pi^u_\beta(z;Q), \qquad \pi^u_\alpha(z;Q) = \sum_\beta M_{\alpha\beta} c^u_\beta(z;Q). \tag{2}$$

Here M is the unique Möbius-inversion matrix of this fixed finite order. The averaged atom is

$$\bar{\pi}^u_\alpha(Q) = \sum_{z \in \mathrm{supp}(Q)} Q(z) \pi^u_\alpha(z; Q). \tag{3}$$

Equations (2)–(3) specify what "Möbius atom" and "averaged atom" mean everywhere below; no unstated vector-index convention is used. The signed-net coordinate is the informative coordinate minus the misinformative coordinate, both before and after inversion.

The proofs treat the following standard results as cited prerequisites rather than reproving them: the strong law of large numbers, the pointwise ergodic theorem, elementary continuity and the mean value theorem for log, nonnegativity of Kullback–Leibler divergence, finite-poset Möbius inversion [10], Hoeffding's inequality, and the Fannes–Audenaert entropy-continuity inequality. Every application records its required finite-alphabet, support, independence, or denominator premise. "Exact real" means arithmetic in the mathematical real numbers; it is deliberately distinct from a theorem about a particular binary64 implementation.

2 Finite setting and shared-exclusions quantities

Fix a source count m, finite source alphabets $\mathcal{S}_1, \ldots, \mathcal{S}_m$, a finite target alphabet $\mathcal{T}$, and the joint alphabet

$$\mathcal{Z} = \mathcal{S}_1 \times \cdots \times \mathcal{S}_m \times \mathcal{T}. \tag{4}$$

Let P be one probability law on $\mathcal{Z}$, and put

$$S = \mathrm{supp}(P), \qquad p_{\min} = \min_{z \in S} P(z) > 0. \tag{5}$$

All logarithms are natural, so all information values use nats. The redundancy lattice is fixed and finite. Its nodes are nonempty antichains of nonempty source subsets. In `pid-rs`, $m \in \{2, 3, 4\}$ for SxPID and $m \in \{2, 3\}$ for $I_{\min}$.

Definition 2.1 (FA-D1: cumulative-prefix empirical law)**.** The label FA-D1 and the phrase "cumulative-prefix empirical law" are local to this document. They name the empirical law of the first n observations in one nested sequence and do not invoke a published theorem.

Let $Z_1, Z_2, \ldots$ be one nested observation sequence. For every integer $n \geq 1$, define the empirical law of its first n rows by

$$\widehat{P}_n(z) = \frac{1}{n} \sum_{j=1}^{n} \mathbf{1}\{Z_j = z\}. \tag{6}$$

Its masses are nonnegative and

$$\sum_{z \in \mathcal{Z}} \widehat{P}_n(z) = \frac{1}{n} \sum_{j=1}^{n} \sum_{z \in \mathcal{Z}} \mathbf{1}\{Z_j = z\} = 1, \tag{7}$$

so it is a probability law. Prefix $n+1$ contains every row from prefix n.

FA-D1 is not a sliding-window law, a resampled law, or a sequence produced after refitting a transform at every n.

Fix a supported realization $z = (s_1, \ldots, s_m, t)$ and a lattice node α. Define

$$A_\alpha(s) = \bigcup_{a \in \alpha} \{S_a = s_a\}, \tag{8}$$

$$B_\alpha(s,t) = A_\alpha(s) \cap \{T = t\}, \tag{9}$$

$$C(t) = \{T = t\}. \tag{10}$$

Each of these three events contains z. Its P-mass is therefore at least $P(z) \geq p_{\min}$. For any law Q whose required masses are positive, the cumulative informative and misinformative terms are

$$c_\alpha^+(z;Q) = -\log Q(A_\alpha(s)), \tag{11}$$

$$c_\alpha^-(z;Q) = \log \frac{Q(C(t))}{Q(B_\alpha(s,t))}. \tag{12}$$

The cumulative net term is $c_\alpha^{\text{sx}} = c_\alpha^+ - c_\alpha^-$. A fixed Möbius matrix M maps each cumulative vector to its pointwise atom vector. The averaged atom under Q is the Q-weighted sum over joint realizations.

The two-source dictionary and a worked empirical example. For $m = 2$, the four antichains are

$$\alpha_{\text{red}} = \{\{1\},\{2\}\}, \qquad \alpha_1 = \{\{1\}\}, \qquad \alpha_2 = \{\{2\}\}, \qquad \alpha_{\text{syn}} = \{\{1,2\}\}.$$

The only covering relations are

$$\alpha_{\text{red}} \prec \alpha_1 \prec \alpha_{\text{syn}}, \qquad \alpha_{\text{red}} \prec \alpha_2 \prec \alpha_{\text{syn}},$$

and α_1, α_2 are incomparable. For the signed-net atoms, these nodes are called redundancy, unique information from S_1, unique information from S_2, and synergy, respectively. For each $u \in \{+,-,\text{sx}\}$, the zeta relation is explicitly

$$c^u_{\alpha_{\text{red}}} = \pi^u_{\alpha_{\text{red}}}, \tag{13}$$

$$c^u_{\alpha_1} = \pi^u_{\alpha_{\text{red}}} + \pi^u_{\alpha_1}, \tag{14}$$

$$c^u_{\alpha_2} = \pi^u_{\alpha_{\text{red}}} + \pi^u_{\alpha_2}, \tag{15}$$

$$c^u_{\alpha_{\text{syn}}} = \pi^u_{\alpha_{\text{red}}} + \pi^u_{\alpha_1} + \pi^u_{\alpha_2} + \pi^u_{\alpha_{\text{syn}}}. \tag{16}$$

As a complete empirical calculation, take the four binary XOR rows

$$(s_1, s_2, t) \in \{(0,0,0), (0,1,1), (1,0,1), (1,1,0)\},$$

each once, and write $Q = \widehat{P}_4$. Thus

$$Q(s_1, s_2, t) = \begin{cases} 1/4, & t = s_1 \oplus s_2, \\ 0, & \text{otherwise.} \end{cases}$$

For every supported row, $Q(C(t)) = 1/2$, and the cumulative values are

Node	$Q(A_\alpha)$	$Q(B_\alpha)$	c_α^+	c_α^-	c_α^{sx}
α_{red}	3/4	1/4	$\log(4/3)$	$\log 2$	$\log(2/3)$
α_1	1/2	1/4	$\log 2$	$\log 2$	0
α_2	1/2	1/4	$\log 2$	$\log 2$	0
α_{syn}	1/4	1/4	$\log 4$	$\log 2$	$\log 2$

Möbius inversion therefore gives

Atom	π^+	π^-	$\pi^{\text{sx}} = \pi^+ - \pi^-$
Redundancy	$\log(4/3)$	$\log 2$	$\log(2/3)$
Unique S_1	$\log(3/2)$	0	$\log(3/2)$
Unique S_2	$\log(3/2)$	0	$\log(3/2)$
Synergy	$\log(4/3)$	0	$\log(4/3)$

All four supported rows give the same pointwise values, so these are also the averaged atoms. They reconstruct

$$\log(2/3) + 2\log(3/2) + \log(4/3) = \log 2 = I_Q((S_1, S_2); T).$$

In particular, the negative redundancy is not clipped: together with either positive unique atom it gives $I_Q(S_i; T) = 0$.

Williams–Beer $I_{\min}$ coordinates. For a nonempty source collection a and a target value with $Q(t) > 0$, define the source-specific information

$$J_a(t; Q) := D\Big(Q_{S_a|T=t} \,\Big\|\, Q_{S_a}\Big) = \sum_{s_a : Q(s_a,t)>0} \frac{Q(s_a, t)}{Q(t)} \log \frac{Q(s_a, t)}{Q(s_a)Q(t)}. \tag{17}$$

For an antichain α, the target-specific minimum and its target average are

$$R_\alpha(t; Q) := \min_{a\in\alpha} J_a(t; Q), \tag{18}$$

$$I_\cap^{\min}(\alpha; Q) := \sum_{t:Q(t)>0} Q(t) R_\alpha(t; Q). \tag{19}$$

The values $I_\cap^{\min}(\alpha; Q)$ are the cumulative Williams–Beer redundancies on the same finite antichain order. Their PID atoms are

$$\pi_\alpha^{\min}(Q) := \sum_\beta M_{\alpha\beta} I_\cap^{\min}(\beta; Q), \tag{20}$$

where M is the Möbius-inversion matrix from Equation (2). Zero-mass target values are omitted. A finite minimum of continuous coordinates remains continuous at a tie, although it need not be differentiable.

3 Deterministic convergence and sampling corollary

Theorem 3.1 (FA-1: Deterministic plug-in implication)**.** *Let Q_n be probability laws on $\mathcal{Z}$. Assume*

(i) $Q_n(z) \to P(z)$ *for every* $z \in \mathcal{Z}$*; and*

(ii) $\operatorname{supp}(Q_n) \subseteq S$ *for every* n*.*

Then there is a finite N_0 such that $\operatorname{supp}(Q_n) = S$ for every $n \geq N_0$. On this tail, all supported-event denominators are positive. The following exact-real quantities converge to their corresponding values under P:

1. *every supported-realization SxPID cumulative informative and misinformative term;*
2. *every informative, misinformative, and signed-net Möbius atom;*
3. *every probability-weighted averaged SxPID atom;*
4. *every supported-target $I_{\min}$ specific-information value and every fixed two- or three-source $I_{\min}$ atom; and*
5. *every fixed finite-alphabet Shannon entropy, mutual information, co-information, and O-information expression formed from the same law.*

Pointwise SxPID convergence is keyed by the joint realization, not by a returned-vector index.

Proof. For each $z \in S$, coordinate convergence and $P(z) > 0$ give $Q_n(z) > P(z)/2$ for all sufficiently large n. The support is finite, so one finite N_0 works for every supported cell. Assumption (ii) excludes new cells. This proves support equality on the tail.

Fix z and α. Event masses are finite linear sums of cell masses, so the masses of A_α, B_α, and C converge. Their limits are at least $P(z) > 0$. Continuity of the logarithm on $(0, \infty)$ gives convergence of c_α^+ and c_α^-. The fixed finite linear map M preserves convergence. Subtraction gives the net atoms. After support stabilization, an average is a fixed finite sum whose weights and pointwise values both converge. No off-support $x \log x$ argument is needed for this SxPID step because Assumption (ii) and eventual support equality keep the average on one fixed finite support face.

For the specific-information values in Equation (17), support stabilization fixes every positive marginal cell and makes every required denominator positive. Each finite term is continuous. A finite minimum of continuous functions is continuous, including at a tie. Target averaging and the fixed Möbius map preserve convergence. A tie can remove differentiability, but it does not remove continuity.

Finally, define $h(0) = 0$ and $h(x) = -x \log x$ for $x > 0$. The function h is continuous on $[0, 1]$. Each finite-alphabet entropy is a finite sum of h. Marginalization and fixed finite linear combinations preserve convergence. □

Corollary 3.2 (FA-2: i.i.d. or stationary ergodic sampling). *The assumptions of Theorem 3.1 hold almost surely when either*

(i) the observations are i.i.d. with law P; or

(ii) the process is strictly stationary and ergodic with one-time marginal law P.

Strict stationarity without ergodicity is not sufficient.

Proof. Apply the strong law, or the pointwise ergodic theorem, to the indicator of each cell. A finite intersection gives simultaneous coordinate convergence. If $P(z) = 0$, every time index has zero probability of that cell. A countable union shows that no such cell occurs, almost surely. Thus, the empirical support is always a subset of S on one probability-one event. □

4 Quantitative local moduli for SxPID

Let p be the probability vector of P, and let q be another probability law on the same finite joint alphabet. Define

$$\delta = \|q - p\|_1 , \qquad L = \log(1/p_{\min}), \tag{21}$$

and assume $\delta \le p_{\min}/2$. For every event E,

$$|q(E) - p(E)| \le \delta. \tag{22}$$

This deliberately loose inequality is sufficient for the following stated constants.

Theorem 4.1 (SxPID local bounds). *For each supported realization and lattice node,*

$$|c_\alpha^+(z; q) - c_\alpha^+(z; p)| \le \frac{2\delta}{p_{\min}}, \tag{23}$$

$$|c_\alpha^-(z; q) - c_\alpha^-(z; p)| \le \frac{4\delta}{p_{\min}}. \tag{24}$$

Let

$$\|M\|_\infty = \max_i \sum_j |M_{ij}|. \tag{25}$$

The pointwise atom errors obey

$$\textit{informative:} \quad |\Delta\pi^+| \le \frac{2\,\|M\|_\infty\,\delta}{p_{\min}}, \tag{26}$$

$$\textit{misinformative:} \quad |\Delta\pi^-| \le \frac{4\,\|M\|_\infty\,\delta}{p_{\min}}, \tag{27}$$

$$\textit{net:} \quad |\Delta\pi^{\mathrm{sx}}| \le \frac{6\,\|M\|_\infty\,\delta}{p_{\min}}. \tag{28}$$

If also $\mathrm{supp}(q) \subseteq S$*, the averaged errors obey*

$$\textit{informative:} \quad |\Delta\bar\pi^+| \le \|M\|_\infty \left(\frac{2}{p_{\min}} + L\right)\delta, \tag{29}$$

$$\textit{misinformative:} \quad |\Delta\bar\pi^-| \le \|M\|_\infty \left(\frac{4}{p_{\min}} + L\right)\delta, \tag{30}$$

$$\textit{net:} \quad |\Delta\bar\pi^{\mathrm{sx}}| \le \|M\|_\infty \left(\frac{6}{p_{\min}} + L\right)\delta. \tag{31}$$

Proof. Each required event E has $p(E) \ge p_{\min}$. Equations (22) and $\delta \le p_{\min}/2$ give $q(E) \ge p_{\min}/2$. The mean-value theorem for log gives

$$|\log q(E) - \log p(E)| \le \frac{|q(E) - p(E)|}{\min\{q(E), p(E)\}} \le \frac{2\delta}{p_{\min}}. \tag{32}$$

The informative term contains one logarithm. The misinformative term contains two. The row-sum norm of M gives the first two atom bounds in (26); the triangle inequality gives the net bound.

For one atom component π, decompose its averaged error as

$$\left|\sum_{z\in S} q(z)\pi(z;q) - \sum_{z\in S} p(z)\pi(z;p)\right| \le \sum_{z\in S} q(z)|\pi(z;q) - \pi(z;p)| \tag{33}$$

$$+ \sum_{z\in S} |q(z) - p(z)|\,|\pi(z;p)|. \tag{34}$$

At p, each cumulative informative and misinformative coordinate lies in $[0, L]$. For the net term, $c_\alpha^+ - c_\alpha^- \in [-L, L]$. Apply the row-sum norm once. This gives one factor of L for each of the three weight-change terms and proves the averaged bounds. □

Remark 4.2 (Why the common-support condition matters)**.** Under $\delta \le p_{\min}/2$, the condition $\mathrm{supp}(q) \subseteq S$ implies $\mathrm{supp}(q) = S$. Without common support, a concrete two-source law shows the obstruction. Let all three alphabets be binary, let

$$P = \delta_{(0,0,0)}, \qquad q_\varepsilon = (1-\varepsilon)\delta_{(0,0,0)} + \varepsilon\delta_{(1,1,1)}, \qquad 0 < \varepsilon < \frac{1}{2}.$$

Then

$$\mathrm{TV}(q_\varepsilon, P) = \varepsilon, \qquad \|q_\varepsilon - P\|_1 = 2\varepsilon.$$

At the bottom node $\alpha_{\mathrm{red}} = \{\{1\},\{2\}\}$, the shared source event for $(0,0,0)$ has mass $1-\varepsilon$, while the event for $(1,1,1)$ has mass ε. The misinformative term is zero at both rows, and the averaged informative—hence also net—bottom atom is

$$\bar\pi^{\mathrm{sx}}_{\alpha_{\mathrm{red}}}(q_\varepsilon) = -(1-\varepsilon)\log(1-\varepsilon) - \varepsilon\log\varepsilon = h_2(\varepsilon),$$

whereas its value under P is zero. Therefore

$$\frac{\left|\bar{\pi}^{\mathrm{sx}}_{\alpha_{\mathrm{red}}}(q_\varepsilon) - \bar{\pi}^{\mathrm{sx}}_{\alpha_{\mathrm{red}}}(P)\right|}{\|q_\varepsilon - P\|_1} = \frac{h_2(\varepsilon)}{2\varepsilon} \longrightarrow \infty.$$

Thus no support-independent linear constant can replace the common-support premise.

These are exact-real law-level inequalities. They do not include binary64 rounding, platform logarithm variation, resource rejection, or implementation refinement. The Rust SxPID and $I_{\min}$ paths reject sample counts above 2^{53}.

5 Secondary $I_{\min}$ modulus

This section is a secondary validation control. It is not the research priority of the document. Assume $\mathrm{supp}(q) \subseteq S$ and $\delta \leq p_{\min}/2$. For a supported target t, let

$$r = q(s_a, t), \qquad u = q(s_a), \qquad v = q(t), \qquad w = \frac{r}{v}, \qquad \ell = \log \frac{r}{uv}. \tag{35}$$

Marginalization contracts joint L^1 distance. Every positive p-marginal is at least $p_{\min}$, and every corresponding q-marginal is at least $p_{\min}/2$. Hence

$$\sum_{s_a} |w_q(s_a) - w_p(s_a)| \leq \frac{4\delta}{p_{\min}}, \tag{36}$$

and

$$|\ell_q - \ell_p| \leq \frac{6\delta}{p_{\min}}. \tag{37}$$

To see (36), put $r_q(s_a) = q(s_a, t)$, $v_q = q(t)$, and write

$$\sum_{s_a} \left| \frac{r_q(s_a)}{v_q} - \frac{r_p(s_a)}{v_p} \right| \leq \frac{\sum_{s_a} |r_q(s_a) - r_p(s_a)|}{v_q} + \frac{|v_q - v_p|}{v_q} \tag{38}$$

$$\leq \frac{2\delta}{p_{\min}/2}. \tag{39}$$

Each of the three logarithm factors changes by at most $2\delta/p_{\min}$. Equation (37) is their sum. Also, $r_p, u_p, v_p \in [p_{\min}, 1]$ whenever $r_p > 0$, and $r_p \leq u_p, v_p$. Hence $p_{\min} \leq r_p/(u_p v_p) \leq 1/p_{\min}$ and $|\ell_p| \leq L \leq \log(2/p_{\min})$. Splitting the specific-information difference gives

$$|J_a(t; q) - J_a(t; p)| \leq \sum_{s_a} w_q(s_a) |\ell_q(s_a) - \ell_p(s_a)| \tag{40}$$

$$+ \sum_{s_a} |w_q(s_a) - w_p(s_a)|\, |\ell_p(s_a)|. \tag{41}$$

The first sum is at most $6\delta/p_{\min}$ because w_q is a probability vector. The second is at most $(4\delta/p_{\min}) \log(2/p_{\min})$ by Equation (36). Therefore,

$$|J_a(t; q) - J_a(t; p)| \leq \left[4 \log\left(\frac{2}{p_{\min}}\right) + 6\right] \frac{\delta}{p_{\min}}. \tag{42}$$

A finite minimum is one-Lipschitz in the sup norm. If C_J is the coefficient of δ in (42), define

$$R_\alpha(t; Q) = \min_{a \in \alpha} J_a(t; Q), \tag{43}$$

$$I_\cap^{\min}(\alpha; Q) = \sum_t Q(t) R_\alpha(t; Q). \tag{44}$$

Each $J_a(t;p)$ is a Kullback–Leibler divergence, so it is nonnegative. Also,

$$J_a(t;p) = \sum_{s_a} p(s_a \mid t) \log \frac{p(t \mid s_a)}{p(t)} \le -\log p(t) \le L. \tag{45}$$

Thus, $0 \le R_\alpha(t;p) \le L$. Marginalization gives $\|q_T - p_T\|_1 \le \delta$. Therefore,

$$\left| I_\cap^{\min}(\alpha;q) - I_\cap^{\min}(\alpha;p) \right| \le \sum_t q(t) |R_\alpha(t;q) - R_\alpha(t;p)| \tag{46}$$

$$+ \sum_t |q(t) - p(t)| R_\alpha(t;p) \tag{47}$$

$$\le (C_J + L)\delta. \tag{48}$$

Since $L = \log(1/p_{\min})$, the last line is the target-averaged redundancy bound. The fixed $I_{\min}$ Möbius matrix multiplies this bound by its absolute row-sum norm. A minimum tie preserves continuity but can remove differentiability and a generic normal limit.

A complete minimum-tie crossing. Let T be a uniform bit. For $|\theta| < 1/4$, let

$$e_1(\theta) = \frac{1}{4} + \theta, \qquad e_2(\theta) = \frac{1}{4} - \theta,$$

let $N_1 \sim \text{Bernoulli}(e_1(\theta))$ and $N_2 \sim \text{Bernoulli}(e_2(\theta))$ be mutually independent and independent of T, and put

$$S_i = T \oplus N_i.$$

Equivalently, with $b_e(0) = 1 - e$ and $b_e(1) = e$, the full-support joint law is

$$P_\theta(s_1, s_2, t) = \frac{1}{2} b_{e_1(\theta)}(s_1 \oplus t)\, b_{e_2(\theta)}(s_2 \oplus t).$$

Each S_i is uniform, and for either target value

$$J_{\{i\}}(t; P_\theta) = f(e_i(\theta)), \qquad f(e) := \log 2 - h_2(e),$$

where $h_2(e) = -e \log e - (1-e)\log(1-e)$. Since

$$f'(e) = \log \frac{e}{1-e} < 0 \quad \text{for } 0 < e < \frac{1}{2},$$

the redundancy node $\alpha_{\text{red}} = \{\{1\},\{2\}\}$ satisfies

$$R_{\alpha_{\text{red}}}(t; P_\theta) = \min\{f(e_1(\theta)), f(e_2(\theta))\} = f\left(\frac{1}{4} + |\theta|\right).$$

The source-specific values tie at $\theta = 0$ and exchange minimizers across zero. Since $f'(1/4) = -\log 3$, the left derivative is $+\log 3$ and the right derivative is $-\log 3$. Target averaging does not change this expression, so the averaged $I_{\min}$ redundancy is continuous but nondifferentiable at the crossing.

6 Shannon and ratio bounds

Definitions and sign conventions. For a finite tuple X_A, write

$$H_Q(X_A) := -\sum_{x_A : Q(x_A) > 0} Q(x_A) \log Q(x_A), \qquad H_Q(\varnothing) = 0.$$

Mutual information and conditional mutual information are the entropy combinations

$$I_Q(X;Y) = H_Q(X) + H_Q(Y) - H_Q(X,Y), \tag{49}$$
$$I_Q(X;Y \mid Z) = H_Q(X,Z) + H_Q(Y,Z) - H_Q(Z) - H_Q(X,Y,Z). \tag{50}$$

The implementation uses Bell's co-information sign:

$$\begin{aligned} \mathrm{CoI}_{\mathrm{Bell}}(X_1,X_2;T) &:= I(X_1;T) + I(X_2;T) - I((X_1,X_2);T) && (51)\\ &= I(X_1;X_2) - I(X_1;X_2 \mid T) && (52)\\ &= H(X_1) + H(X_2) + H(T) - H(X_1,X_2) - H(X_1,T) \\ &\quad - H(X_2,T) + H(X_1,X_2,T). && (53) \end{aligned}$$

The McGill-style interaction-information convention recorded by this project has the opposite sign:

$$\mathrm{II}_{\mathrm{McGill}} = -\,\mathrm{CoI}_{\mathrm{Bell}}\,.$$

Thus the Bell-sign co-information of two independent uniform bits and their XOR target is $-\log 2$.

For $r \geq 2$, with X_{-i} denoting all variables except X_i, the O-information is

$$\Omega(X_1,\ldots,X_r) := (r-2)H(X_1,\ldots,X_r) + \sum_{i=1}^{r} H(X_i) - \sum_{i=1}^{r} H(X_{-i}).$$

For $r = 3$, this equals the Bell-sign co-information. Positive O-information is conventionally called redundancy-dominated and negative O-information synergy-dominated; it is not itself a PID atom.

The project-defined target-free ratios are

$$\mathrm{Red}^{\circ}(X_1,\ldots,X_r) := \frac{\sum_i H(X_i)}{H(X_1,\ldots,X_r)}, \tag{54}$$
$$\begin{aligned} \mathrm{Vul}^{\circ}(X_1,\ldots,X_r) &:= \frac{\sum_i H(X_i \mid X_{-i})}{H(X_1,\ldots,X_r)} && (55)\\ &= \frac{rH(X_1,\ldots,X_r) - \sum_i H(X_{-i})}{H(X_1,\ldots,X_r)}. && (56) \end{aligned}$$

They require positive joint entropy. The published target-conditioned ratios are

$$\bar{r}(T;S_1,\ldots,S_m) := \frac{\sum_i I(T;S_i)}{I(T;S_1,\ldots,S_m)}, \tag{57}$$
$$\begin{aligned} \bar{v}(T;S_1,\ldots,S_m) &:= \frac{\sum_i I(T;S_i \mid S_{-i})}{I(T;S_1,\ldots,S_m)} && (58)\\ &= \frac{\sum_i \left[I(T;S_1,\ldots,S_m) - I(T;S_{-i})\right]}{I(T;S_1,\ldots,S_m)}. && (59) \end{aligned}$$

Here $I(T;\varnothing) = 0$. These ratios require positive joint mutual information. All four ratios are unitless. Any reporting threshold is a software policy, not part of the mathematical definitions.

Let one entropy term have alphabet size $K \geq 2$, and put

$$\varepsilon = \tfrac{1}{2}\left\|q - p\right\|_1. \tag{60}$$

Apply Audenaert's bound [1] to diagonal density matrices with diagonals p and q. Their trace distance is ε, and their von Neumann entropies are $H(p)$ and $H(q)$. For $0 \leq \varepsilon \leq 1 - 1/K$, the Fannes–Audenaert bound gives

$$|H(q) - H(p)| \leq g_K(\varepsilon) := \varepsilon \log(K-1) + h_2(\varepsilon), \tag{61}$$

where

$$h_2(x) = -x \log x - (1-x)\log(1-x), \qquad h_2(0) = h_2(1) = 0. \tag{62}$$

The function g_K is increasing only up to $1 - 1/K$. For a marginal A with alphabet size $K_A \geq 2$ and only a joint variation bound available, the safe bound is

$$|H(q_A) - H(p_A)| \leq g_{K_A}(\min\{\varepsilon, 1 - 1/K_A\}). \tag{63}$$

If the marginal variation ε_A itself is available and $\varepsilon_A \leq 1 - 1/K_A$, use $g_{K_A}(\varepsilon_A)$. If $\varepsilon_A > 1 - 1/K_A$, use the valid cap $\log K_A$. If $K_A = 1$, the entropy difference is zero. Marginalization gives $\varepsilon_A \leq \varepsilon$, which is why Equation (63) is safe. A fixed entropy linear combination is bounded by the sum of the absolute coefficients times the term-specific bounds.

Why clipping is necessary. Direct substitution of the joint variation can fail. On the four cells $(A, B) = (0,0), (0,1), (1,0), (1,1)$, take

$$p = (1, 0, 0, 0), \qquad q = \left(\frac{3}{10}, \frac{1}{5}, \frac{1}{2}, 0\right).$$

Then

$$\mathrm{TV}(p, q) = \frac{1}{2}\left(\frac{7}{10} + \frac{1}{5} + \frac{1}{2}\right) = \frac{7}{10},$$

whereas

$$p_A = (1, 0), \qquad q_A = (1/2, 1/2), \qquad \mathrm{TV}(p_A, q_A) = \frac{1}{2}.$$

Consequently,

$$|H(q_A) - H(p_A)| = \log 2.$$

For a binary alphabet, $g_2(x) = h_2(x)$, and direct un-clipped substitution would give

$$g_2(0.7) = h_2(0.7) \approx 0.610864 < \log 2 \approx 0.693147,$$

which is not an upper bound. Clipping at $1 - 1/2 = 1/2$ instead gives $g_2(1/2) = \log 2$, exactly as required.

Lemma 6.1 (Ratio perturbation). *Let $D_p \geq d > 0$, $|D_q - D_p| \leq d/2$, and $|N_p| \leq B$. Then*

$$\left|\frac{N_q}{D_q} - \frac{N_p}{D_p}\right| \leq \frac{2|N_q - N_p|}{d} + \frac{2B|D_q - D_p|}{d^2}. \tag{64}$$

Proof. The denominator condition gives $D_q \geq d/2$. Decompose the difference into $(N_q - N_p)/D_q + N_p(D_p - D_q)/(D_q D_p)$ and apply the triangle inequality. □

Target-free normalized limits require positive population joint entropy. Target-conditioned normalized limits require positive population joint mutual information. Every target-conditioned input term must describe the same law, target, source order, units, preprocessing, evaluation sample, and estimand. For the convergence claim, each input sequence must be the corresponding exact-real finite-alphabet plug-in term formed from the same empirical law. A positive denominator alone does not establish this coherence.

For a report threshold τ, a population denominator strictly above τ guarantees eventual `Defined` status. A denominator below τ guarantees an eventual status other than `Defined`. Equality gives no eventual status guarantee.

7 A time-uniform i.i.d. envelope

This section uses Hoeffding's published inequality [6] and union bounds. The labeled composition is project documentation; no mathematical priority or scientific-novelty claim is made.

Theorem 7.1 (FA-3: time-uniform i.i.d. cumulative-prefix envelope)**.** *Assume the FA-D1 sequence is i.i.d. from one fixed K-cell law, with $K \geq 1$. Let $0 < \alpha < 1$ and*

$$\alpha_n = \frac{6\alpha}{\pi^2 n^2}. \tag{65}$$

With probability at least $1 - \alpha$, simultaneously for every $n \geq 1$,

$$\left\| \widehat{P}_n - P \right\|_1 \leq B_n := K\sqrt{\frac{\log(K\pi^2 n^2/(3\alpha))}{2n}}. \tag{66}$$

Proof. For one fixed cell z and one fixed n, Hoeffding's inequality gives

$$\mathbb{P}(|\widehat{P}_n(z) - P(z)| \geq x) \leq 2e^{-2nx^2}. \tag{67}$$

Set

$$x_n = \sqrt{\frac{\log(K\pi^2 n^2/(3\alpha))}{2n}}. \tag{68}$$

Since $2K\exp(-2nx_n^2) = \alpha_n$, the failure probability for one cell at time n is at most α_n/K. A union bound over K cells and all n costs

$$\sum_{n\geq 1} \alpha_n = \frac{6\alpha}{\pi^2} \sum_{n\geq 1} \frac{1}{n^2} = \alpha. \tag{69}$$

On the complement, summing the K coordinate bounds gives (66). □

Define

$$N_0 = \min\{r \geq 1 : B_n \leq p_{\min}/2 \text{ for every } n \geq r\}. \tag{70}$$

The integer N_0 is finite because $B_n \to 0$. Intersect the concentration event with the probability-one support-containment event of Corollary 3.2. On the intersection, support equality holds for all $n \geq N_0$. Substitute B_n for the actual δ in Theorem 4.1.

For a fixed prefix N, a separate support-discovery bound is

$$\mathbb{P}(\operatorname{supp}(\widehat{P}_N) \neq S) \leq |S|e^{-Np_{\min}} \leq Ke^{-Np_{\min}}. \tag{71}$$

Indeed, one supported cell z is missing after N i.i.d. rows with probability $(1 - P(z))^N \leq e^{-NP(z)} \leq e^{-Np_{\min}}$; a union bound over S gives the first inequality. Once a state enters a cumulative prefix, it remains in every later prefix. This persistence does not hold for a sliding window. A usable N_0 needs a known positive lower bound on $p_{\min}$. An empirical minimum cannot replace it because an unobserved rare state can have positive population mass. The simultaneous L^1 bound can be read at an almost-surely finite data-dependent index. Support and local-bound conclusions additionally need that index to be at least N_0. The law must remain the same fixed i.i.d. law. The statement does not cover sliding or otherwise dependent windows, drift, feedback that changes the sampling law, rejection-selected samples, or stationary ergodic data without an additional rate assumption.

The theorem controls an exact-real empirical law. It is not a confidence sequence for Rust output. It does not include floating-point error, the platform logarithm, resource failure, dependence, drift, rejection selection, or the 2^{53} sample-count limit.

8 Independent frozen transforms

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be the underlying probability space and let $\mathcal{G} \subseteq \mathcal{F}$ be generated by the training artifact. Let $\mathcal{X}$ be standard Borel, let $\mathcal{Z}$ be finite, and let $\perp \notin \mathcal{Z}$. Suppose

$$\Phi : \Omega \times \mathcal{X} \longrightarrow \mathcal{Z} \cup \{\perp\}$$

is $\mathcal{G} \otimes \mathcal{B}(\mathcal{X})$-measurable. The same frozen map $\Phi_\omega(x) := \Phi(\omega, x)$ is used for every evaluation row and every prefix.

Let $W_1, W_2, \ldots$ be conditionally i.i.d. given $\mathcal{G}$, with common regular conditional law κ_ω on $\mathcal{X}$. An i.i.d. raw sequence independent of the training artifact is the special case in which κ_ω is a fixed law. Put

$$Y_j(\omega) := \Phi(\omega, W_j(\omega))$$

and assume

$$\kappa_\omega(\{x : \Phi_\omega(x) = \perp\}) = 0 \quad \text{almost surely.} \tag{72}$$

Corollary 8.1 (Conditional push-forward target)**.** *Define*

$$P_\Phi(\omega; z) := \int_{\mathcal{X}} \mathbf{1}\{\Phi_\omega(x) = z\}\, \kappa_\omega(dx), \qquad z \in \mathcal{Z}.$$

On an event of probability one,

$$\widehat{P}_n^{\Phi}(z) := \frac{1}{n} \sum_{j=1}^{n} \mathbf{1}\{Y_j = z\} \longrightarrow P_\Phi(z) \qquad \text{for every } z \in \mathcal{Z}.$$

The empirical support is always contained in $\operatorname{supp}(P_\Phi)$ *and equals it for all sufficiently large* n*. Hence Theorem 3.1 applies pathwise with population law* P_Φ*. The limit is generally not the same functional evaluated at the unconditional mixture* $\bar{P}_\Phi(z) := \mathbb{E}[P_\Phi(z)]$*.*

Proof. Conditional i.i.d. means that, for every k and $z_1, \ldots, z_k \in \mathcal{Z}$,

$$\mathbb{P}(Y_1 = z_1, \ldots, Y_k = z_k \mid \mathcal{G})(\omega) \tag{73}$$

$$= \int_{\mathcal{X}^k} \prod_{j=1}^{k} \mathbf{1}\{\Phi_\omega(x_j) = z_j\}\, \kappa_\omega^{\otimes k}(dx_{1:k}) = \prod_{j=1}^{k} P_\Phi(\omega; z_j). \tag{74}$$

Thus, on almost every conditioning fiber, the transformed sequence is i.i.d. with the fixed finite law $P_\Phi(\omega; \cdot)$. Equation (72) and a countable union ensure that every transformed row lies in $\mathcal{Z}$.

The ordinary strong law on each product-law fiber gives simultaneous convergence of the finitely many cell indicators. A zero-probability cell is never observed on that fiber, while convergence to a positive cell probability makes its empirical frequency positive eventually. This proves support containment and stabilization. Applying Theorem 3.1 fiberwise and integrating the resulting conditional probability-one event proves the claim. ☐

An `OutOfRangePolicy::Error` transform satisfies the corollary only when the conditional evaluation law has mass one inside every inclusive fitted range and satisfies every input-validity rule. If it violates the zero-failure assumption, let $\eta = \eta(\mathcal{G})$ be its conditional per-row failure probability. For a training outcome with $\eta > 0$, the first n rows all succeed with probability $(1-\eta)^n$, and the infinite prefix fails with conditional probability one. Unconditionally, this failure mechanism has probability $\mathbb{P}(\eta > 0)$. It has probability one only when $\eta > 0$ almost surely.

Conditioning a complete prefix on success does not preserve the original-law claim. Explicit rejection sampling can produce a sequence that is conditionally i.i.d. given $\mathcal{G}$ from a conditional law when success has positive probability. That sequence has a different estimand and needs a declared sampling contract. `OutOfRangePolicy::ClampToBoundary` supplies a total range map for otherwise valid finite inputs, but it defines a tail-clamped categorical estimand. It is not the error-policy estimand.

Same-row fitting, changing transforms, arbitrary pooling across different fitted maps, and target-adaptive fitting on evaluation rows are outside the corollary. Training-target-adaptive partial least squares is compatible only when fitted on the independent training artifact, frozen, and applied to disjoint conditionally i.i.d. evaluation rows.

9 Counterexamples and invalid stronger claims

Lens	Construction	Consequence
Support face	Add a new cell of mass $\varepsilon \downarrow 0$.	A local informative term can grow as $-\log \varepsilon$; uniform linear averaged bounds fail without common support.
Probability	Draw $C \sim \mathrm{Bernoulli}(1/2)$ and set $Z_j = C$ for every j.	The process is stationary but not ergodic; a path sees only one state.
Rare state	Give one state mass ε.	It is absent after n rows with probability $(1-\varepsilon)^n$; no uniform deterministic discovery time exists.
$I_{\min}$ tie	Cross a law where two specific-information values exchange order.	The minimum is continuous but need not be differentiable; a generic delta-method normal limit does not follow.
Transform	Fit finite min/max ranges, then evaluate an independent standard-normal held-out law with an error policy.	Every finite range leaves positive out-of-range mass; infinite-prefix success has probability zero for each positive-failure training outcome.
Mixture	Randomly choose a correlated or anticorrelated binary push-forward law.	Each conditional law has mutual information $\log 2$, but their equal mixture is independent and has mutual information zero.
Indexing	Append a rare realization that is lexicographically first.	Returned vector positions shift; pointwise convergence requires realization keys.
Numerics	Exceed exact binary64 integer counts or change the platform logarithm.	The exact-real theorem does not become a portable executable theorem.
Downstream	Reuse overlapping windows, drift the law, or select complete samples after inspecting them.	The i.i.d. proof and its envelope do not apply.

The following stronger claims are invalid and are retained for audit:

1. Strict stationarity is sufficient without ergodicity. The latent-constant construction in the table disproves it; ergodicity or another pathwise frequency premise is required.
2. All-unique samples identify population support. A finite sample cannot prove that no unobserved population cell has positive mass.
3. Pointwise atoms converge by returned-vector index. A lexicographically earlier state shifts later positions, so convergence is keyed by the realization.
4. One deterministic support-discovery time works for every finite law. A cell of arbitrarily small positive mass rules out a uniform time.
5. Complete-success conditioning preserves the original error-policy estimand. It does not; explicit rejection sampling can define a different conditional-law estimand only under its own declared sampling contract.
6. The conditional fitted-transform limit equals the PID of the unconditional mixture. Nonlinearity of PID in the law rules this out in general.
7. Foldwise PMFs or atoms can be pooled when each fold uses a different map. No triangular-array or cross-fit theorem establishing that operation is proved here.
8. The Lean artifact proves the stochastic theorem or the Rust implementation. It proves only the deterministic exact-real lemmas listed in Section 10.
9. A fixture tolerance is a portable binary64 error theorem. The tolerance applies only to the committed cases and arithmetic path.
10. Unrestricted exact sign and tie evaluation is always practical after denominator clearing. A rational linear combination of logarithms can reduce in principle to an integer-product comparison, but the required powers can be too large for an unrestricted practical implementation; no such API exists here.
11. The all-prefix i.i.d. bound covers sliding or dependent windows. It does not; a valid dependence theorem and its premises are required.
12. Continuity at an $I_{\min}$ tie implies a normal limit. Continuity alone supplies neither differentiability nor a central limit theorem.

10 Formal and executable evidence boundary

The Lean 4.33 root project's deterministic fixed-support core proves these exact-real lemmas:

- eventual positivity at a positive limit;
- convergence of a fixed finite event mass;
- positivity of a finite sum of nonnegative cell masses when the event contains a positive cell;
- sequential-limit preservation by log and $-\log$ at positive limits, and by log ratios at positive limits;
- sequential-limit preservation by $-x\log x$, including at zero;

- preservation of convergence by fixed finite linear and weighted sums;
- preservation of convergence by a finite nonempty minimum, including ties; and
- preservation by a finite weighted sum of finite positive-limit log-ratio combinations.

Separate modules imported by the same root project formalize the finite heterogeneous keyed Sx event layer: source, target, and target-restricted events; their equivalence-class covers; anchor and intersection properties; positive mass under positive anchor mass; and the law-independent event map. A further imported module proves a generic finite equivalence-union fractional-cover bound and its source, target-restricted, and target-event corollaries. The `TwoSourceCountEventBridge.lean` module additionally defines an exact empirical law from a supplied natural count function with positive total and proves the event-count, positivity, local logarithm, and positive-support average identities for the source-one, source-two, joint-source, and redundancy signed-net cumulative nodes.

The `TwoSourceMobiusAtomBridge.lean` module then fixes the categorical two-source sub-scope's four cumulative nodes, four concrete Möbius atoms, three components, and 24-coordinate order. It proves inverse Möbius/zeta algebra, commutation with empirical averaging, all-coordinate supplied-count equalities, exact rational and real products, and scaled-log sign and zero equivalences. The paper-facing formulas are a reviewed repository transcription, not a formal publication-to-Lean correspondence theorem.

Lean does not connect bytes or rows to that count function, encode stochastic limit theorems, the published component-atom nonnegativity theorem, $I_{\min}$, Rust or floating-point refinement, certifier/parser execution, the support-change transfer between laws, a higher-source lattice, or statistical, population, calibration, or consumer validity. The checker binds the complete Lake manifest and all nine package revisions. It verifies each dependency checkout's root, revision, origin, and clean status with global and system Git configuration and Git environment routing disabled. It retains the checkout's local configuration to verify the recorded origin. The checked project sources cannot contain the tokens `admit`, `axiom`, `constant`, `native_decide`, `sorry`, or `sorryAx`. The checker builds the project and replays its declarations with Lean's bundled kernel checker. It enforces an exact ordered inventory of all 339 source-level declarations across the eight imported modules and audits the permitted axiom basis of all 246 named source theorems. The complete two-source count/event and count-to-atom bridges are separately SHA-256 bound. Two separately digest-pinned semantic contracts fix paper-facing event/count facts and an asymmetric exact 24-coordinate witness; compiled examples are not counted as named-theorem axiom audits. The self-test uses baseline-first static and isolated Lean changes under normal and optimized Python to exercise imports, theorem strength, order, components, counts, weights, products, signs, scope, and the semantic contracts. CI runs the same formal checker. This does not enlarge the theorem boundary beyond supplied exact counts and the fixed two-source categorical surface.

The independent generator `scripts/generate-finite-alphabet-plugin-oracle.py` uses 100-digit standard-library Decimal arithmetic and direct definitions; it imports no `pid-rs` code. Its digest-bound fixture covers all 4, 18, and 166 averaged SxPID coordinates in the listed two-, three-, and four-source tables, informative, misinformative, and signed-net values, realization-key changes after a late rare state, listed two- and three-source $I_{\min}$ tables, and left/tie/right minimizer crossings. The companion Rust test, rather than the JSON fixture, separately checks bit-identical outputs from fitted-quantizer wrappers and direct calls on the same categorical matrices, and it checks pointwise omission of an absent realization on the listed support face. The committed Decimal comparisons use the absolute envelope

$$64\,\texttt{f64::EPSILON}\ \text{nats}. \tag{75}$$

This is bounded implementation evidence. It is not the convergence proof, external review, population validation, or a global floating-point bound.

11 Downstream contract boundary

Consumer	Covered path	Uncovered path
Prisoma	Use conditionally i.i.d. cases or episodes from one common conditional law. Use one evaluation row per unit, or separately establish the row law. Fit and freeze transforms on an independent partition.	Independent non-identical units, dependent rows within an episode, same-row fitting, pooling outputs from different maps, or other dependence without a theorem.
Galadriel	Use direct categorical SxPID on i.i.d. rows with a fixed alphabet.	Continuous KSG on dependent temporal windows, same-window scaling, added-noise studies, circular delete-block work, or complete-sample rejection after a degeneracy check.
Haldir	Use i.i.d. or strictly stationary and ergodic fixed-law rows for finite-alphabet entropy, co-information, O-information, SxPID, and positive-denominator ratio limits.	A fixed one-time marginal without ergodicity, drift, CUSUM, alarms, overlapping windows, post-alarm estimation, or calibration.
Crebain	Use stable categorical and fitted-transform contracts after an explicit dependency and input-contract review.	No compatibility or adapter claim follows from this theorem.

No downstream repository is a runtime dependency of `pid-rs`. This audit states mathematical preconditions. It does not claim downstream qualification.

12 Reproduction

Run these commands from the repository root:

```
python3 scripts/generate-finite-alphabet-plugin-oracle.py
cargo test --locked -p pid-core --test finite_alphabet_plugin_oracle
python3 scripts/check-lean-finite-convergence.py
scripts/check-finite-alphabet-convergence-pdf.sh
python3 scripts/check-method-catalog.py
```

13 Correction ledger

The review process corrected the following defects in the current reviewed draft:

1. A local-bound draft reused $p_{\min}$ from a law different from the local reference law. The final statement sets $p = P$.

2. A Shannon draft substituted joint variation into a nonmonotone marginal Fannes–Audenaert envelope. Equation (63) now clips at $1 - 1/K_A$.
3. A transform draft did not require the random map to be training-measurable. A map such as $Q(\omega, x) = x$ xor $W_1(\omega)$ can otherwise destroy conditional independence.
4. A theorem draft omitted the support-stabilized tail and allowed an empty antichain.
5. A stopping statement omitted the condition that support conclusions need an index at least N_0.
6. A failure statement treated a random η as positive almost surely. The correct unconditional probability is $\mathbb{P}(\eta > 0)$.
7. A rejection statement did not distinguish complete-prefix conditioning from a newly declared rejection-sampling estimand.
8. A status statement incorrectly made strict threshold separation necessary for every eventual `Defined` sequence. At threshold zero, the nested count path $(n_{00}, n_{01}, n_{10}, n_{11}) = (2, 0, 0, 2)$ followed by repeated additions $01, 10, 00, 11$ converges to independence while empirical mutual information stays positive.
9. A downstream draft omitted identical-row-law and ergodicity conditions.
10. An evidence draft attributed quantizer inputs to the Decimal JSON. They occur only in the companion Rust test.
11. A valid first averaged-net bound used $2L$. Applying the net cumulative range directly gives the tighter L term in Theorem 4.1.
12. An intermediate entropy draft asserted $p_{\min} \le 1/K$. This is false when K is the declared alphabet size and P has sparse support; one-cell support has $p_{\min} = 1$. No accepted bound uses this claim. Alphabet size and supported-cell mass remain separate.

14 Open Wibral-lineage work

The next research priority is shared-exclusions PID, not an extension of $I_{\min}$. Open work is:

1. a dependence-aware SxPID result for generic mixing or temporal processes that do not admit the explicit independence coloring in the separate result under a dependency coloring;
2. a sequential contract that separates fixed-law monitoring from drift and post-alarm estimation;
3. a triangular-array or cross-fit theorem for changing fitted maps;
4. practical certified sign and tie evaluation for empirical Sx expressions;
5. a deductive refinement to binary64 or interval arithmetic;
6. a Lean encoding of the empirical-law stochastic step and complete three-and-higher-source SxPID lattice beyond the fixed two-source supplied-count bridge, including its Möbius, component, cumulative, atom, and averaging composition; and
7. mathematically valid multivariate and mixed-dimensional extensions needed by ecosystem consumers.

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